
FINAL PRESENTATION

Copula-Guided Graph-Conditioned Synthetic Data Generator

Final Technical Presentation

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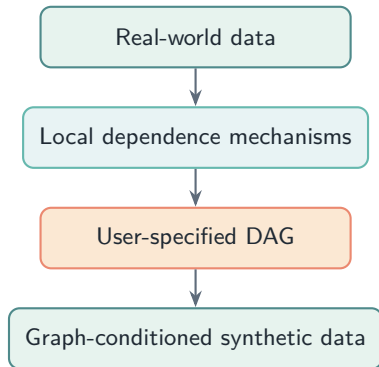
continuous data → local copula mechanisms → graph-conditioned simulation

Problem Statement

GOAL

Generate realistic continuous data from a user-specified graph.

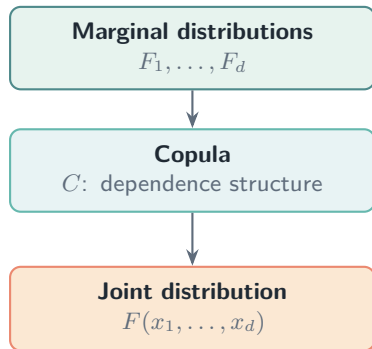
- ▶ Learn local dependence patterns from real data.
- ▶ Reuse them as node-wise generative mechanisms.
- ▶ Preserve the parent structure imposed by the DAG.



Standard generators learn one fixed joint distribution; here the graph tells us how to reassemble reusable local mechanisms.

Copulas: Separating Marginals from Dependence

TWO COMPONENTS OF A JOINT DISTRIBUTION



SKLAR'S THEOREM

$$F(x_1, \dots, x_d) = C(F_1(x_1), \dots, F_d(x_d))$$

On the probability scale,

$$U_j = F_j(X_j), \quad U_j \sim \text{Uniform}(0, 1).$$

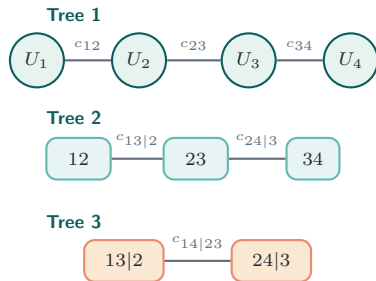
Hence the copula is the joint distribution of

$$(U_1, \dots, U_d).$$

The simulator keeps empirical marginals and fits dependence mechanisms on the U -scale.

Vine Copulas: Building a Copula from Trees

D-VINE: FOUR-VARIABLE STRUCTURE



Edges of Tree k become nodes of Tree $k+1$. Each edge is one bivariate copula selected by BIC.

CONDITIONAL COORDINATES

$$h_{j|k}(u_j, u_k) = \frac{\partial C_{jk}(u_j, u_k)}{\partial u_k} = F_{j|k}(u_j | u_k)$$

$$u_{1|2} = h_{1|2}(u_1, u_2), \quad u_{3|2} = h_{3|2}(u_3, u_2)$$

THREE-PARENT EXTENSION

Three parents and one child form a four-dimensional local mechanism. The selected D-vine order determines the inverse h-function chain used during generation.

How Copulas Enter the Simulator

Each node is sampled conditionally on its parents. The node's indegree determines which local generative mechanism is used.

0 PARENTS



Empirical marginal

$$U_X \sim \text{Unif}(0, 1), \quad X = \widehat{F}_X^{-1}(U_X)$$

1 PARENT



Pair copula

Conditional sampling via the fitted inverse h -function

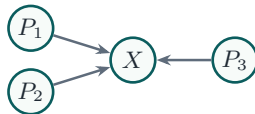
2 PARENTS



Three-variable vine

Sample from the fitted conditional law $X \sim F_{X|P_1, P_2}$

3 PARENTS



Four-variable D-vine

All six parent orderings are fitted and scored

Datasets Used in the Study

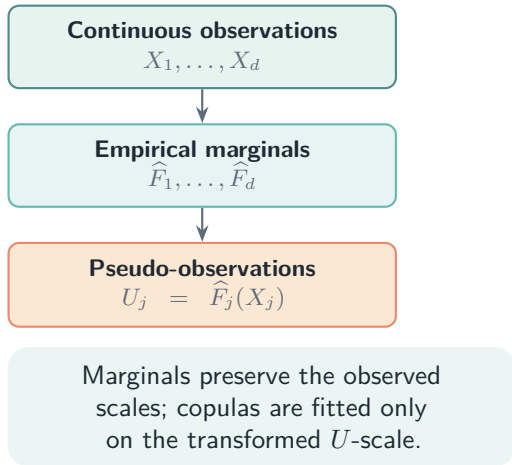
Three continuous datasets from different domains are used to fit, validate and demonstrate the simulator.

Dataset	Source	Shape	Variables used
Sachs	Sachs protein-signalling dataset	853×11	Continuous measurements of 11 phosphorylated proteins and phospholipids
Diabetes	<code>sklearn.datasets</code>	442×10	Age, BMI, blood pressure and six blood-serum measurements
Breast Cancer	<code>sklearn.datasets</code>	569×12	Selected morphology predictors from the mean, error and worst-value feature groups

All three datasets use the same configuration-driven calibration and generation pipeline.

Calibration: From Real Data to Local Mechanisms

STEP 1: MOVE TO THE COPULA SCALE



STEP 2: FIT LOCAL DEPENDENCE MODELS

Candidate families:

- ▶ Independence, Gaussian, Student t
- ▶ Clayton, Gumbel, Frank
- ▶ Rotations allowed

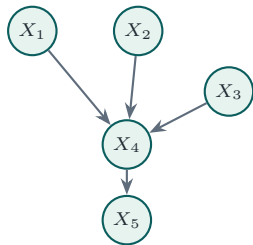
Selection and conditioning:

- ▶ lowest BIC for pair-copula family selection
- ▶ held-out conditional log-likelihood for three-parent order selection
- ▶ adaptive quantile-bin conditional strategy available in final configs

Output of calibration: empirical marginals + reusable local copula libraries.

Generation Under a User-Specified DAG

EXAMPLE WITH THREE PARENTS



topological traversal: parents before child

The DAG is supplied externally and determines which saved local mechanism is used at each node.

GENERATION RULES

0 parents: sample from empirical marginal

1 parent: pair-copula inverse h-function

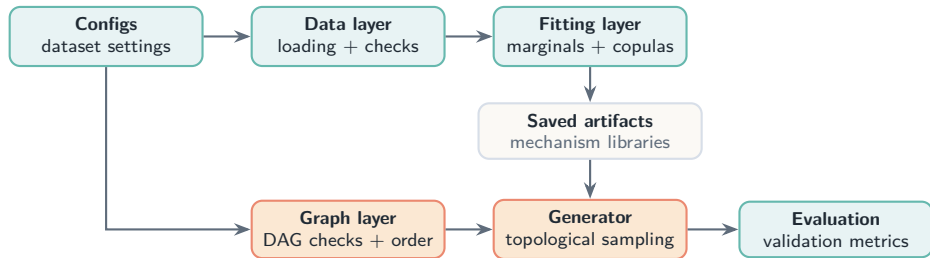
2 parents: three-variable conditional vine

3 parents: selected four-variable D-vine

Final step: inverse empirical marginal maps U -samples back to original scale

Calibration learns reusable pieces; the DAG assembles them into a synthetic dataset.

Software Implementation



Reproducibility

YAML configs, fixed seeds, serialized artifacts and Git-LFS archives.

Validation checks

Graph logic, serialization, generation, h-function inverses and release checks.

Current stable implementation: continuous data with maximum indegree three.

Exhaustive Indegree-Three Libraries

Every possible child with three parents was fitted and serialized for each dataset.

Dataset	Variables	Mechanisms
Sachs	11	1320
Diabetes	10	840
Breast Cancer	12	1980
Total		4140

Why exhaustive precomputation?

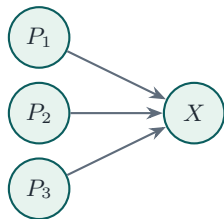
- ▶ generation from any admissible DAG can reuse saved mechanisms;
- ▶ parent sets are looked up by the graph at generation time;
- ▶ artifacts are packaged for reuse through Git LFS.

Mechanism count formula: $d \binom{d-1}{3}$. This is why parallel fitting and reusable artifacts matter.

Synthetic Validation Design

A controlled four-variable mechanism tests whether the fitter recovers the known parent order and joint dependence.

KNOWN GENERATING MECHANISM



All six candidate parent orders are scored using held-out conditional log-likelihood.

FINAL VALIDATION GRID

Sample sizes	500, 1000, 2000
Seeds	five seeds per sample size
Runs	15 total
Metrics	order recovery, score margin, Kendall τ MAE, energy distance

Note: $n = 250$ was excluded because the validation procedure requires at least 400 observations.

Synthetic Validation Results

Values are mean $\pm$ standard deviation over five seeds. Lower Kendall MAE and energy distance are better.

n	Order recovery	Score margin	Kendall τ MAE	Energy distance
500	100%	0.1163 ± 0.0241	0.01029 ± 0.00393	0.001773 ± 0.001124
1000	100%	0.0654 ± 0.0158	0.00530 ± 0.00206	0.000587 ± 0.000082
2000	100%	0.0937 ± 0.0373	0.00549 ± 0.00344	0.000420 ± 0.000089

15/15

generating order recovered

0.00703

mean Kendall τ MAE

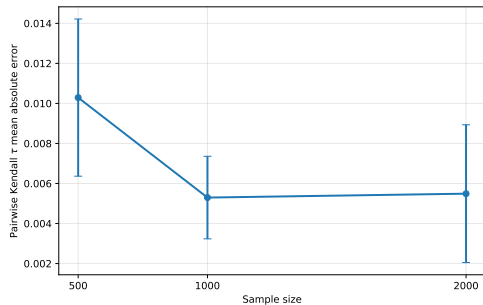
100%

finite and in $[0, 1]$

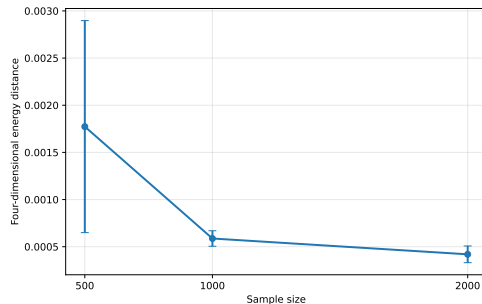
The controlled experiment validates order selection and four-dimensional dependence reconstruction.

Validation Curves: Dependence and Joint Fidelity

RANK-DEPENDENCE ERROR

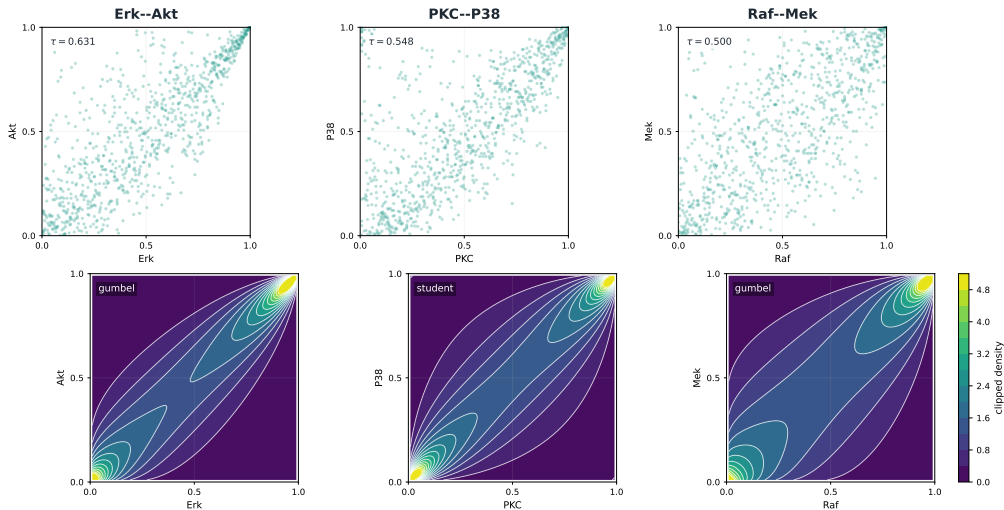


FOUR-DIMENSIONAL JOINT ERROR



Kendall error drops after $n = 500$; energy distance decreases steadily across the validation grid.

What the Fitted Copulas Actually Look Like



Top row: empirical copula-scale observations. Bottom row: BIC-selected fitted pair-copula densities for the same pairs.

Final Scope and Verified Limitations

FINAL STABLE FUNCTIONALITY

- ▶ Continuous tabular datasets.
- ▶ Empirical marginal modeling.
- ▶ Pair-copula one-parent mechanisms.
- ▶ Conditional D-vines for two and three parents.
- ▶ Flexible order selection for three-parent mechanisms.
- ▶ Graph-conditioned generation up to maximum indegree three.
- ▶ Packaged reusable precomputed artifacts.

LIMITATIONS

- ▶ Continuous variables only.
- ▶ Graphs are conditioned on, not inferred.
- ▶ Evaluation focuses on observed distributional fidelity; interventions are outside the current scope.
- ▶ Exhaustive precomputation grows combinatorially with indegree.
- ▶ Conditional-copula modelling remains an approximation.

Conclusion

WHAT WAS BUILT

- ▶ A configuration-driven copula simulator for continuous data.
- ▶ Reusable local mechanisms for 0, 1, 2 and 3 parents.
- ▶ Exhaustive fitted libraries for three datasets.
- ▶ End-to-end graph-conditioned generation from supplied DAGs.
- ▶ Synthetic validation of three-parent order selection and dependence fidelity.

FINAL MESSAGE

Learns local dependence mechanisms from real continuous data



Stores them as reusable copula libraries



Composes them under user-specified DAGs up to indegree three

Thank you